Parameterized and Non-Parameterized Models

ML

1. **Parameterized Machine Learning Functions**

Parameterized models (or functions) are those that have a fixed number of parameters that are learned from the data.

Once the parameters are learned, the model's complexity does not change even if you add more training data.

Model	Parameters
Linear Regression	weights (β_0, β_1, β_2, …)
Logistic Regression	weights and bias
Neural Networks	weights and biases (fixed number once architecture is chosen)
SVM (with linear kernel)	weight vector and bias

✅ **Advantages:**

- Simpler and faster to train
- Easy to interpret
- Require less data

❌ **Disadvantages:**

- Limited flexibility — can't capture complex patterns easily
- Risk of **underfitting**

2. **Non-Parameterized Machine Learning Functions**

Non-parameterized models don't assume a fixed number of parameters.

The model complexity grows with the data size — it learns more structure as more data becomes available.

Model	Explanation
k-Nearest Neighbors (k-NN)	Keeps all data points and compares distances for prediction
Decision Trees	Number of splits (parameters) depends on data

Model	Explanation
Random Forests	Ensemble of trees — complexity grows with data
Non-parametric SVM (with RBF kernel)	Implicitly uses all data through kernel trick

✅ **Advantages:**

- Flexible and can capture complex patterns
- No need to assume a data distribution

❌ **Disadvantages:**

- Computationally expensive
- Need more data and memory
- Harder to interpret

Summary Table

Feature	Parameterized Models	Non-Parameterized Models
Parameters	Fixed number	Grows with data
Model Form	Assumed (e.g., linear)	Data-driven
Flexibility	Low	High
Examples	Linear Regression, Logistic Regression, Neural Networks	k-NN, Decision Trees, Random Forests
Training Speed	Fast	Slower
Memory Requirement	Low	High